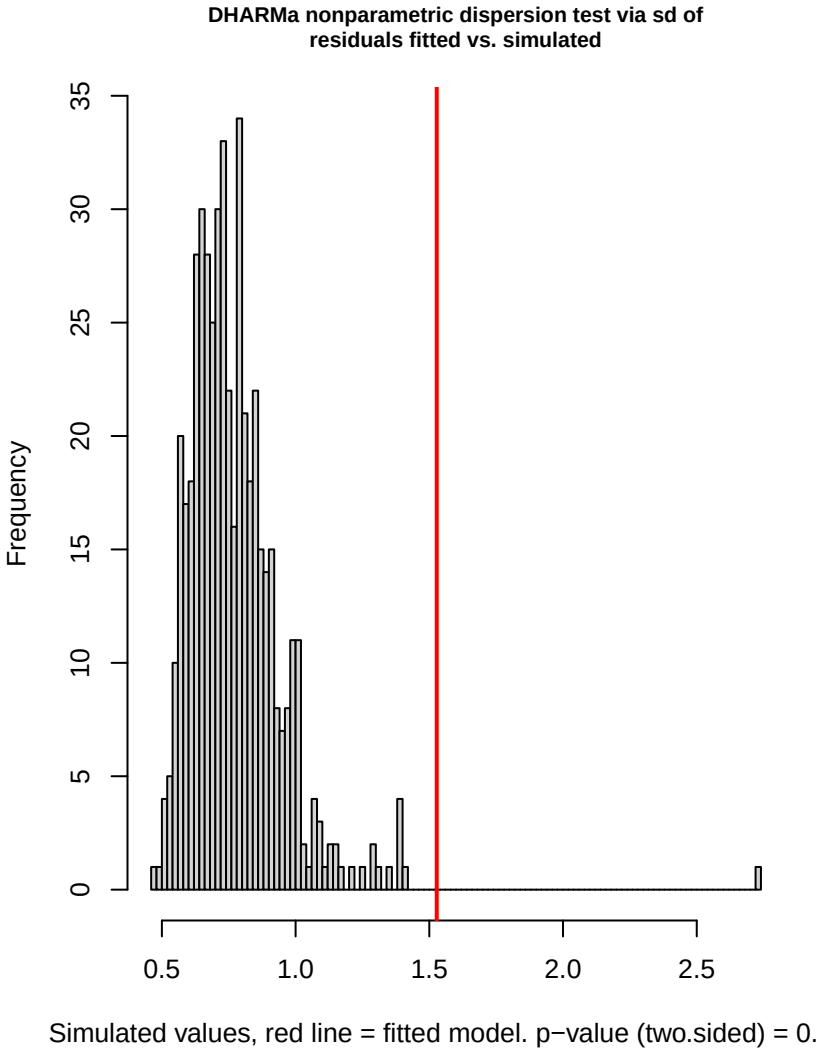
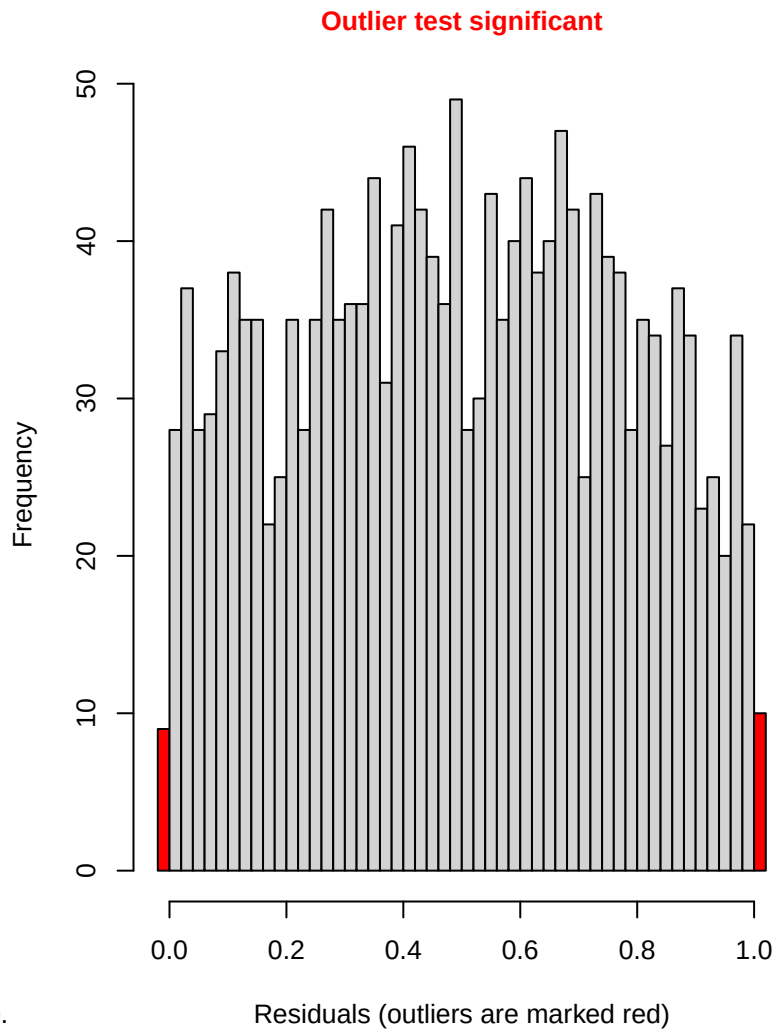
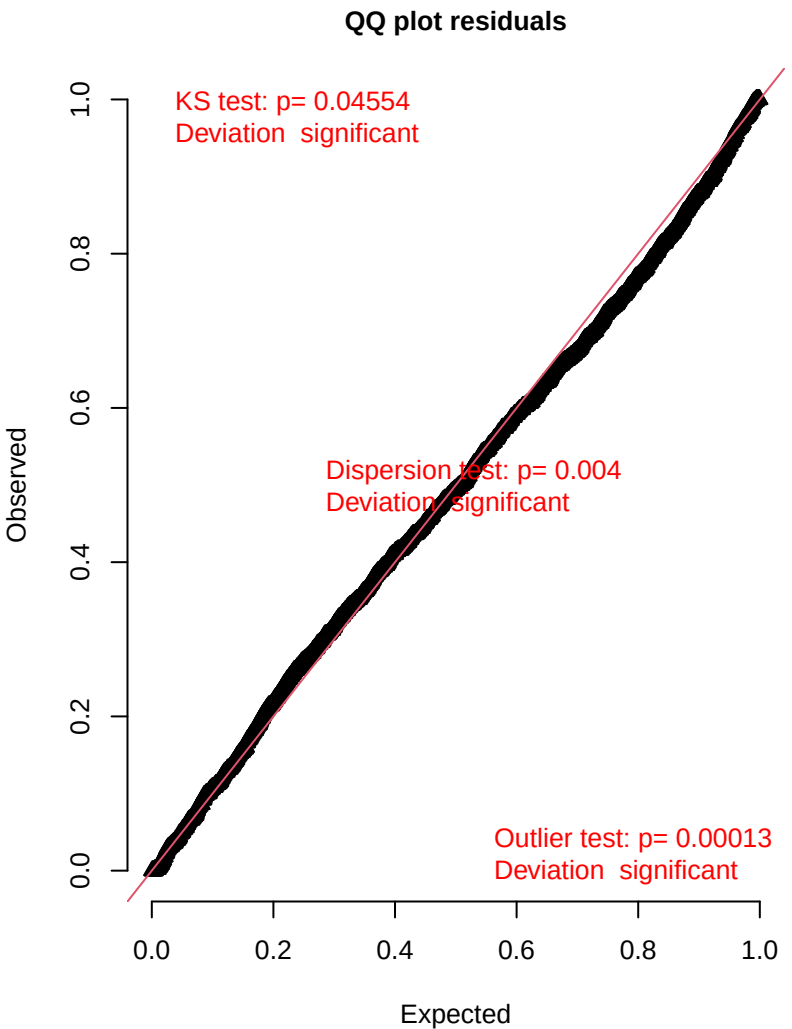
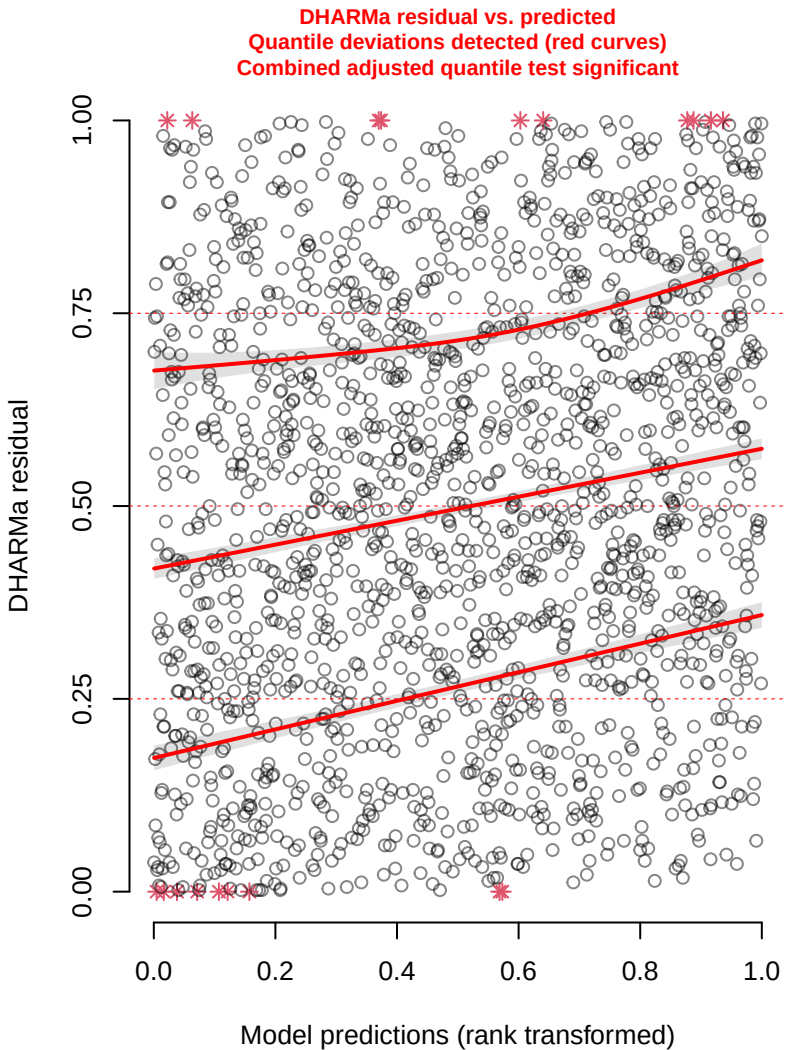
Model validation report

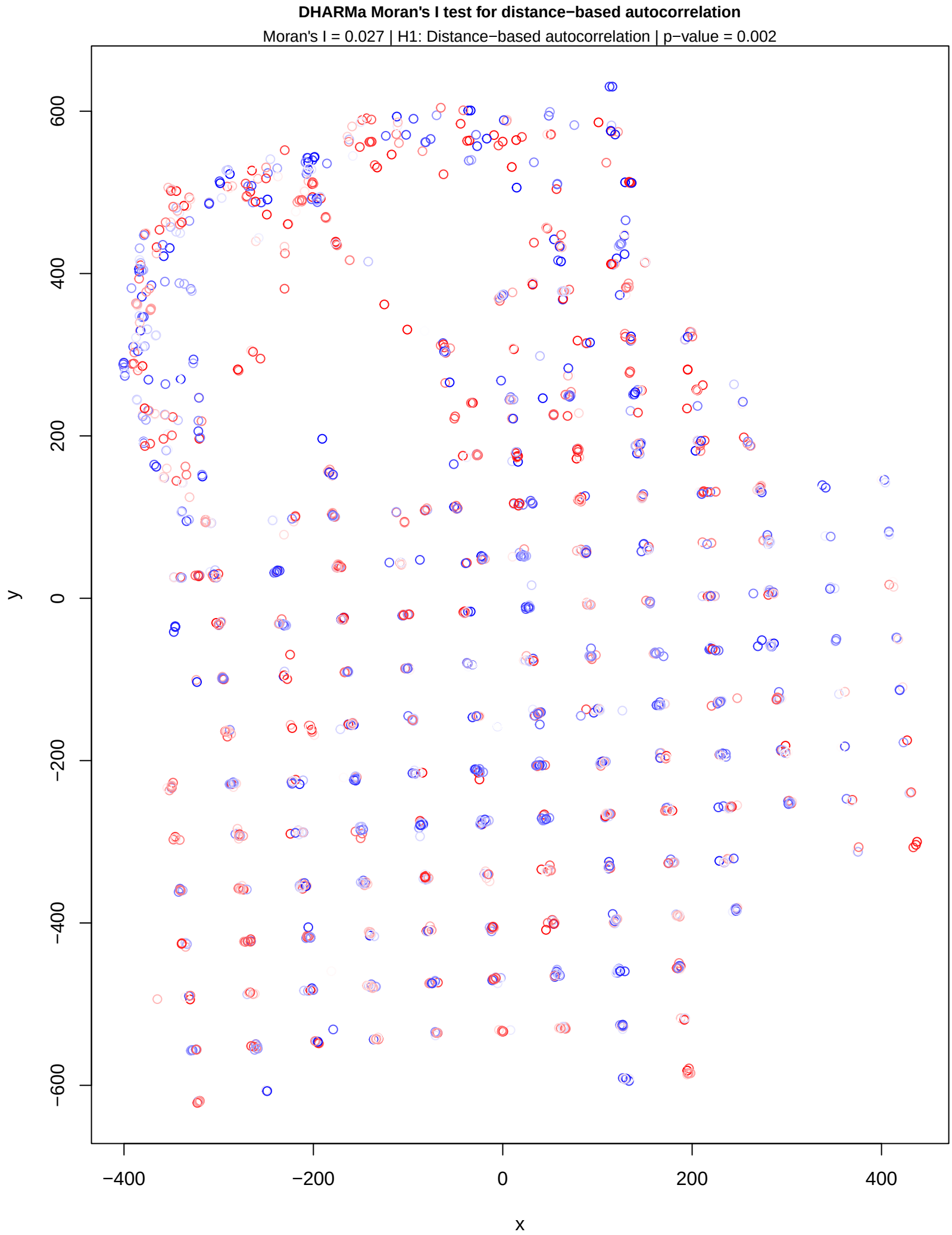
Model: fishing_biomass
Observations: 1755
Date: 2026−04−07

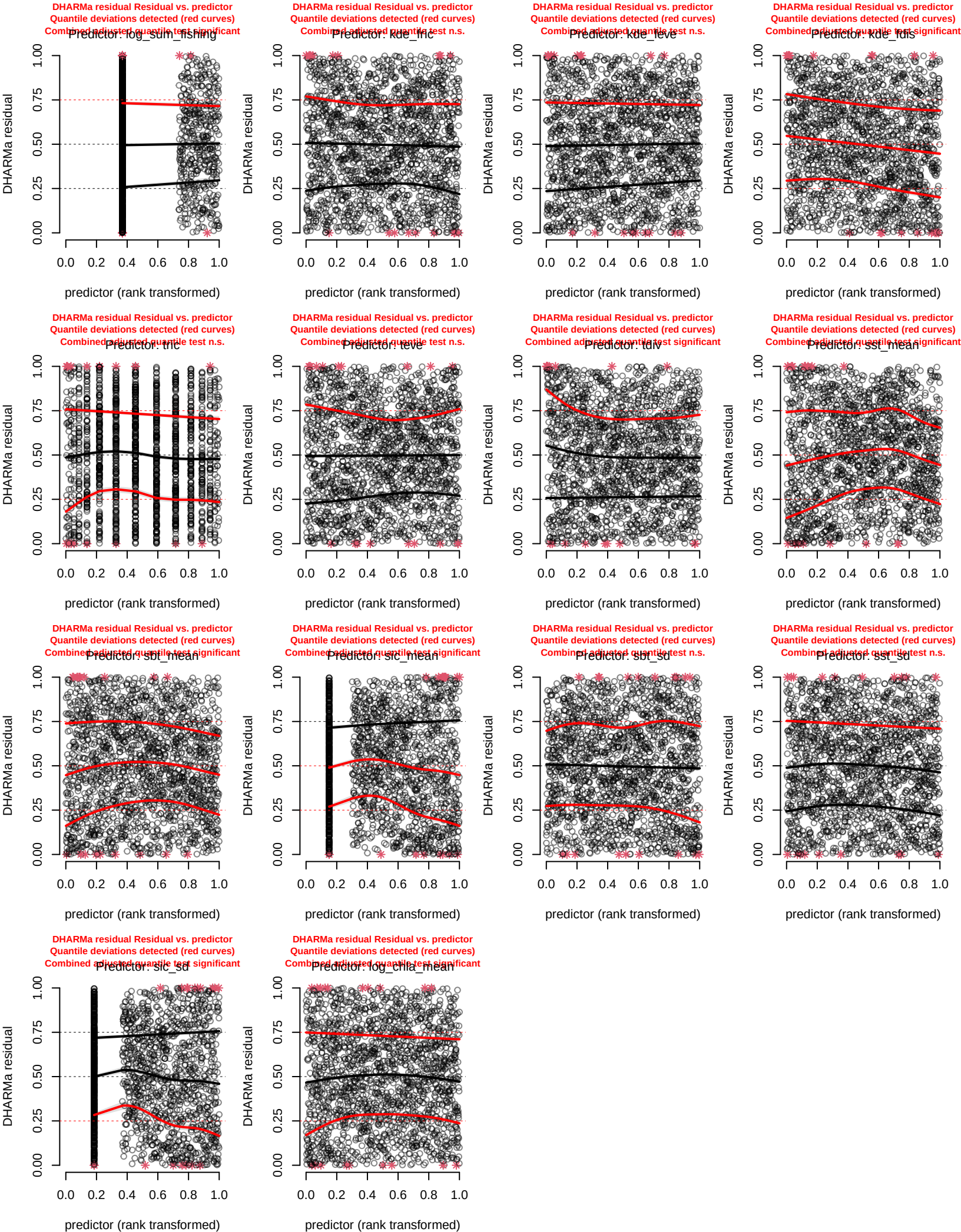

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
20
25
30
35
0.5
1.0
1.5
2.0
2.5
Simulated values, red line = fitted model. p−value (two.sided) = 0.


Outlier test significant
Frequency
0
10
20
30
40
50
0.0
0.2
0.4
0.6
0.8
1.0
Residuals (outliers are marked red)


QQ plot residuals
KS test: p= 0.04554
Deviation significant
Dispersion test: p= 0.004
Deviation significant
Outlier test: p= 0.00013
Deviation significant
Observed
0.0
0.2
0.4
0.6
0.8
1.0
Expected


DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.027 | H1: Distance−based autocorrelation | p−value = 0.002
y
x
600
400
200
0
−200
−400
−600
−400
−200
0
200
400


DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
Predictor: log_sum_fishing
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: kde_tric
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: kde_teve
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
Predictor: kde_tdiv
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: tric
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: teve
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
Predictor: tdiv
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
Predictor: sst_mean
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
Predictor: sbt_mean
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
Predictor: slc_mean
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: sbt_sd
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: sst_sd
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
Predictor: slc_sd
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0